

Group Members

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1. Discuss the consideration / preparation should be taken when performing integration test.
 - Define clear integration test objectives and scope
 - Ensure all individual modules pass unit testing
 - Prepare integration test cases covering interfaces and interactions
 - Setup a proper integration test environment
 - Use appropriate stubs and drivers if needed
 - Plan for error handling and debugging
2. What are the advantages and disadvantages of incremental and non-incremental approaches? Which approach is better? Justify your answer.

Aspect	Incremental Approach	Non-Incremental Approach
Advantages	- Early detection of errors. - Faster delivery of functional parts - Easier to manage and modify - Customer feedback can be incorporated	- Simple to plan and execute - Suitable for smaller projects with stable requirements
Disadvantages	- Requires continuous integration - Higher complexity in managing dependencies - May require additional documentation and coordination	- High risk of failure if errors are found late - debugging is more difficult - Delayed feedback and late issue detection

The Incremental Approach is generally better because it provides early feedback, reduces risk, and allows for gradual improvements. It is especially suitable for large and complex projects where requirements may change over time. The Non-Incremental Approach might work for smaller, well-defined projects but is riskier for larger systems/

3. Compare Unit Test with Integration Test.

Aspect	Unit Test	Integration Test
Scope	Test individual components or units of code in isolation	Test the interaction between different units or components of the system
Purpose	Verifies that each unit of the software performs as expected.	Ensures that combined parts of the application work together as expected.
Speed	Faster to execute	Slower to execute
Maintenance	Easier to maintain due to isolated nature	More complex to maintain due to multiple components and dependencies.

4. In some cases, Integration Test is performed but Unit Test is ignored. Explain the reasons.

Integration testing is sometimes prioritized over unit testing due to factors like tight deadlines, reliance on external systems, and legacy code that lacks modularity. Some teams believe integration tests indirectly validate unit correctness, while others struggle with dependency isolation or lack proper unit testing knowledge. In certain cases, business requirements and system-level behavior take precedence over individual code units. However, neglecting unit tests can make debugging harder and slow development, so a balanced approach incorporating both unit and integration testing is ideal for maintaining software quality.

5. Differentiate Regression Test with Re-Test.

Feature	Regression Testing	Re-Testing
Purpose	Ensures that recent code changes do not impact existing functionalities.	Verifies that a specific defect has been fixed.
Scope	Covers multiple test cases, including both modified and unmodified parts of the application	Focuses only on the failed test cases from previous testing
Execution	Performed after bug fixes, enhancements, or system updates	Performed after a defect has been fixed.

6. What is the difference between Driver and Stub?

Aspect	Driver	Stub
Definition	A driver is a temporary module that simulates the behavior of a calling module.	A stub is a temporary module that simulates the behavior of a called module.
Purpose	Used to test the functionality of a lower-level module when the higher-level module is not available.	Used to test the functionality of a higher-level module when the lower-level module is not available.
Usage Context	Typically used in bottom-up integration testing	Typically used in top-down integration testing .

7. Give an example of an artifact that to be tested/reviewed for each stage in a V – Model. Discuss what the quality attributes are (e.g. Correctness, Consistency, Reliability, Security ...etc.) to be tested for the artifacts.

In the V-Model, artifacts are tested at each stage to ensure quality. **Requirements Analysis** reviews the **SRS** for **correctness and completeness**, while **System Design** evaluates the **architecture** for **scalability and security**. **High-Level and Low-Level Design** focus on **modularity, reusability, and consistency**. **Coding** ensures **functionality and reliability**, followed by **Unit and Integration Testing** for **fault isolation and interoperability**. **System and UAT Testing** validate **performance, usability, and compliance**, ensuring a high-quality final product.